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Predicting chronic kidney disease progression using small pathology datasets and explainable machine learning models

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ABSTRACT

Background: Chronic kidney disease (CKD) poses a major global public health burden, with over 700 million affected. Early identification of those in whom the disease is likely to progress enables timely therapeutic interventions to delay advancement to kidney failure.

Methods: This study developed explainable machine learning models leveraging pathology data to accurately predict CKD trajectory, targeting improved prognostic capability even in early stages using limited datasets. Key variables used in this study include age, gender, most recent estimated glomerular filtration rate (eGFR), mean eGFR, and eGFR slope over time prior to the incidence of kidney failure. Supervised classification modelling techniques included decision tree and random forest algorithms selected for interpretability. Internal validation on an Australian tertiary centre cohort ($n = 706$; 353 with kidney failure and 353 without) achieved exceptional predictive accuracy. To address the inherent class imbalance, centroid-cluster-based under-sampling was applied to the Australian dataset. For external validation, the model was applied to a dataset ($n = 597$ adults) sourced from a Japanese CKD registry. Transfer learning was subsequently employed by fine-tuning machine learning models on 15 % of the external dataset ($n = 89$) before evaluating the remaining 508 patients.

Results: Internal validation achieved exceptional predictive accuracy, with the area under the receiver operating characteristic curve (ROC-AUC) reaching 0.94 and 0.98 on the binary task of predicting kidney failure for decision tree and random forest, respectively. External validation demonstrated performant results with an ROC-AUC of 0.88 for the decision tree and 0.93 for the random forest model. Decision tree model analysis revealed the most recent eGFR and eGFR slope as the most informative variables for prediction in the Japanese cohort.

Conclusion: The research highlights the utility of deploying explainable machine learning techniques to forecast CKD trajectory even in the early stages utilising limited real-world datasets.

Introduction

Over 700 million people were living with chronic kidney disease (CKD) globally in 2017 [1]. CKD is defined as a reduction in the kidney filtration rate and/or the presence of proteinuria for 3 months or more. CKD is traditionally classified according to stages, with Stage 1/2 being early-stage, Stage 3/4 mid-stage and Stage 5 kidney failure. In Australia, there are an estimated 1.7 million aged 18 and over with CKD [2,3]. Whilst most people with CKD are living in early (63.9 %) and mid (34.6 %) stage kidney disease, there has been a more than doubling of people requiring kidney replacement therapy (in the form of dialysis or transplantation) between 2000 – 2020 to over 27,000 Australians, which is predicted to increase a further 30 % by 2030 [4]. The early identification of people who are likely to progress to kidney failure presents an opportunity to deliver targeted interventions to alter the trajectory of the disease. Indeed, early detection and management can prevent progressive CKD by as much as 50 %, a figure that may be even higher in the era of new renoprotective therapies [5,6]. The healthcare, societal and

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economic implications are profound if kidney failure can be delayed or prevented. The annual cost of CKD was estimated to be $9.9 billion in Australia in 2021, with upwards of $1.2 billion absorbed by dialysis alone [7]. For the individual, time without kidney failure is associated with a higher quality of life, and maintenance of day-to-day activities including the ability to work.

CKD is diagnosed with a combination of a blood test to assess biochemical kidney function and a urine test to determine the amount of protein in the urine [3,8]. The data arising is used to determine the Stage of CKD categorizing the condition from stage 1 (mild damage) to stage 5 (kidney failure), which is essential for determining the appropriate management and treatment strategies. Despite the simplicity of these investigations, proteinuria data is often missing, compromising optimal care delivery [9]. Furthermore, the vast number of factors that may influence disease progression are not captured with these two tests alone. Currently, predicting who is likely to progress to kidney failure is based on clinical acumen underpinned by the trajectory of kidney disease. There is a clinical need to be able to identify those individuals who are at greatest risk of progressive disease and kidney failure by integrating other relevant data sets.

To assist with early identification, various statistical predictors have been developed. Kidney Failure Risk Equation (KFRE) is a risk prediction formula that estimates the risk of progression to kidney failure for an individual with chronic kidney disease (CKD) at 2 and 5 years [10]. It has been internationally validated, including in Australia, and is in clinical use in some jurisdictions [11]. Essential variables in the equation include age, sex, eGFR, and urine albumin/creatinine ratio. Serum albumin, bicarbonate, phosphorus, and adjusted calcium are optional parameters. However, the equation has important limitations in that it requires a urine albumin/creatinine ratio, which is infrequently done in clinical practice, applies only to later stages of CKD (G3-G5) and considers only kidney failure (requiring dialysis) as the outcome [9]. Furthermore, the KFRE is a static equation which does not take multiple readings and variation of variables over a time into account.

Recent advancements in artificial intelligence (AI) and machine learning have shown promise in improving the prediction of CKD progression compared to traditional risk equations like the KFRE [12-14]. AI models can analyse complex, longitudinal data and uncover non-linear relationships between variables, potentially leading to more accurate and personalised risk predictions [8,15]. Ferguson and colleagues developed a random forest model to predict the progression of chronic kidney disease (CKD) using patient demographics and over 80 different laboratory variables, such as complete blood cell counts, chemistry panels, and urine analysis [16]. This retrospective study aimed to predict a 40 % decline in eGFR or kidney failure using data from a single time point. The model achieved a high area under the curve (AUC) in both internal testing (0.88 at 2 years and 0.84 at 5 years) and external validation (0.87 at 2 years and 0.84 at 5 years). Aoki and colleagues developed a machine learning model applicable across all stages of CKD [12]. Their retrospective observational study conducted using laboratory data involved 110,264 adults over 5 years. The model utilised random forest survival methods with a 7-variable risk classifier. It accurately predicted a significant decline in eGFR (>30 %) within 5 years, with an AUC-ROC of 0.85. Remarkably, it performed consistently across diverse demographic groups, with an AUC range of 0.83–0.88, and relied on standard pathology results, enhancing its applicability and cost-efficiency. The model highlighted the eGFR slope as the most significant predictor of progression, a novel insight compared to prior studies. Other key variables included initial eGFR, urine albumin-to-creatinine ratio, serum albumin, age, and sex.

Previous studies on CKD prediction and prognosis using ML techniques have emphasized the impact of dataset characteristics and variable selection on model performance. However, in these studies, there is mostly a lack of external validation on independent datasets, which is crucial for assessing generalizability. The use of neural networks as a baseline model is also unexplored. Many previous models for CKD prognosis have a "black box" nature, hindering clinical adoption due to poor interpretability. XAI techniques like SHapley Additive explanations (SHAP) and Local Interpretable Model-agnostic Explanations (LIME) improve model transparency and trustworthiness, bridging the gap between ML predictions and clinical applicability. In addition to XAI techniques, counterfactual analysis emerges as a pivotal method, slightly modifying input data points to observe changes in the model's predictions and providing tangible and actionable insights into the model's decision-making process. To address the limitations of previous studies we developed an explainable longitudinal machine learning model that could effectively identify CKD patients who would ultimately progress to kidney failure and validated the model using an external dataset.

Methodology

In this section, the proposed approach to forecast the CKD trajectory using both the Australian and an external validation dataset is presented. We initially undertook a retrospective analysis utilising existing pathology data that was routinely stored in the electronic medical record systems at the Department of Pathology at Northern Health in Melbourne, Australia over the 5-year period spanning from January 1st, 2019, to December 31st, 2023. Upon receiving ethics approval after thorough procedures to ensure patient privacy and anonymity were in place, we were provided secured access to approximately 10,000 eligible patient records that met the key inclusion criteria of being adult patients aged 18 years and over at the time of diagnosis with a baseline estimated glomerular filtration rate (eGFR) of $\geq$15 mL/min/1.73m^2 and $\leq$ 59 mL/min/1.73m^2. Kidney failure was defined when two eGFR readings of <15 ml/min/1,73m^2 were recorded. Biochemistry arising from dialysis centres were excluded. The electronically extracted records contained basic non-identifiable patient details such as age and gender that were routinely stored for standard clinical pathology purposes and allowed the research team to assign a unique de-identified study number to each patient to further uphold confidentiality principles. Additionally, the Northern Health co-investigator on our study team ensured that all relevant data points were properly anonymised by removing any potential personal identifiers before recording select information into a secure spreadsheet. The secure spreadsheet was developed a priori specifically for the purposes of this analysis based on the identification of key variables and data fields required to answer our research question on predictive modelling from a thorough review of similar previously published studies in leading nephrology journals. The study obtained ethics approval from the Austin Health Human Research Ethics Committee: HREC/100,996/Austin-2023.

Data preparation

The Australian dataset was collected internally, with 149,491 patients being assessed as eligible. To validate the model performance, a Japanese dataset consisting of 1138 newly visiting stage G2–G5 CKD patients was utilised [17]. The dataset comprised demographic, biochemical, and clinical data of patients identified with CKD (eGFR values between 15 and 59 mL/min/1.73m^2). The distinctive characteristics included gender and age, eGFR calculated using the 2009 CKD Epidemiology Collaboration (CKD-EPI) creatinine equation without the race-based coefficient, Serum/LDL/HDL cholesterol, triglycerides, albumin-adjusted calcium, potassium, sodium, phosphate, ferritin, creatinine, urate, C-reactive protein (CRP), 25-hydroxyvitamin D, haemoglobin and haemoglobin A1c (HbA1c) and Urinary albumin-to-creatinine ratio, urinary protein-to-creatinine ratio. Follow-up duration data was also extracted. A minimum of three eGFR measurements per subject was required. For the Australian cohort, patients who had two eGFR readings on the same day after their first eGFR reading falling below 15 mL/min/1.73m^2 were considered to be on dialysis and excluded. The final number of patients was 10,064 and 597 for the Australian and Japanese cohorts, respectively. The threshold for defining kidney failure was set for patients who exhibited at least two eGFR readings below 15 mL/min/1.73m^2 with an interval of at least 30 days. In the context of the Japanese dataset, these readings were inherently spaced six months apart. Consequently, the final cohorts comprised 10,064 patients (353 with kidney failure) in the Australian cohort and 597 patients (162 with kidney failure) in the Japanese cohort.

For the Japanese Dataset, the age at each eGFR measurement was adjusted to the nearest whole number based on the exact timing of the test reading since only the age at the initial test was provided. Notably, the eGFR values in the Japanese cohort were initially calculated using the Modification of Diet in Renal Disease (MDRD) study equation, while the Australian data utilised the Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI) equation for eGFR calculation. The eGFR was therefore re-computed for the Japanese Dataset to ensure consistency with the Australian Dataset. Starting from the MDRD-based eGFR, the serum creatinine (SCr) concentrations were reverse-calculated, factoring in the updated age and gender of the patients. The derived SCr values were then converted from the units of mg/dL to μmol/L. Using these standardised SCr values, along with the patient's age and gender, the eGFR was recalculated according to the CKD-EPI formula. Additionally, the categorical variable for gender was encoded. Fig. 1 outlines the patient selection protocol from Australian and Japanese cohorts for CKD prediction.

A comparison of the Australian and Japanese cohorts is provided in Table 1.

Feature engineering

The patient time-series records were truncated prior to the onset of kidney failure, marked by eGFR failing below 15 mL/min/1.73m^2 for prediction. For tree-based models, a linear regression slope of eGFR values over time was obtained for each patient to capture longitudinal trends in kidney function. Subsequently, the selected features included age, gender, the most recent eGFR value, eGFR time progression slope, and mean eGFR prior to kidney failure onset. The features were normalised with standard scaling for both the Australian and Japanese cohorts.

Table 1
Summary statistics for Australian (before under-sampling) and Japanese CKD datasets.

Variables	AU Dataset	JP Dataset
Total number of patients	10,064	597
Total number of records	154,516	3388
Total number of patients with KF	353	162
Total number of records with KF	5543	456
Sex (M/F)	5064/5000	435/162
Age (years)	77.54 ± 12.02	71.55 ± 11.47
Kidney Failure (%)	3.51 %	27.14 %
eGFR_slope (mL/min/1.73 m^2/day)	0.48 ± 3.83 Mean follow up time over 1 year	−0.01 ± 0.02 Mean follow up time over 2 years
eGFR_mean (ml/min/ 1.73m^2)	46.64 ± 15.71	36.62 ± 13.56

Under-sampling (Imbalanced dataset)

The Australian dataset underwent a pre-processing step where centroid-cluster-based under-sampling was applied to address the class imbalance issue, explicitly aiming to balance the majority class (non-kidney failure cases) with the minority class (kidney failure cases). In datasets where one class significantly outweighs the other, models may exhibit a bias towards predicting the majority class, which can result in poor generalisation for instances of the minority class. Centroid-cluster-based under-sampling addresses class imbalance in datasets by reducing the number of instances in the majority class while preserving the data's underlying distribution. The first step involves clustering the instances of the majority class using the K-means algorithm which partitions the majority class data into 'K' clusters based on feature similarity. Each cluster is represented by a centroid, which is the mean of all the points in that cluster. The centroid serves as a representative of the entire cluster. Instead of retaining all instances from the majority class, only the centroids of the clusters are selected as representative samples. The number of clusters (K) is set to match the number of instances in the minority class, ensuring a balanced dataset. For example, if the minority class has 353 instances, the majority class is clustered into 353 clusters, and the 353 centroids are used as the representative samples. By selecting

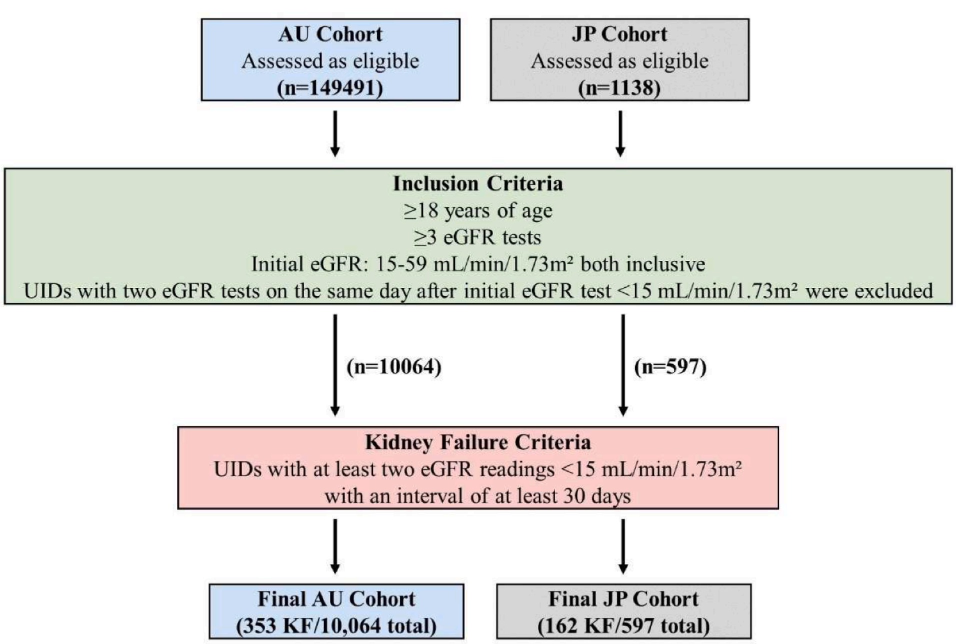

Fig. 1. Data preparation for CKD prediction for Australian and Japanese patient cohorts.

centroids, the method reduces the majority class's size without losing its overall distribution characteristics. This balanced dataset (with an equal number of instances from both classes) is then used for training the ML model. This methodology has been utilised in small-scale medical datasets, such as a study by Zhang et al. [7], which utilised cluster-based under-sampling to address the diagnosis of breast cancer. Similarly, in a study by Jian et al. [8], this methodology was utilised to address the inherent imbalance challenges in predicting eight different diabetes complications. This approach helped improve the model's performance by preventing it from being biased towards the majority class and enhancing its ability to generalize to minority class instances.

Conversely, for the neural network models, specifically Recurrent Neural Networks (RNNs), the slope or mean of the eGFR values were not utilised. Instead, the data was grouped by the patient after normalising the numerical age and eGFR values with min-max scaling and then zero-padded to a fixed sequence length of 50, maintaining consistency before feeding into the RNN. To balance the dataset, TimeSeries KMeans clustering with Dynamic Time Warping (DTW) was applied to the sequences of the majority class, dividing them into clusters based on the similarity of their time series patterns [8]. Instead of selecting representative sequences from each cluster, the centroids of these clusters—synthetic sequences that represent the "average" pattern within each cluster—are used. The number of clusters was chosen to match the number of instances in the minority class, ensuring that the resulting set of centroid sequences from the majority class equals the size of the minority class. These centroid sequences are then combined with the original sequences from the minority class to create a balanced dataset. The sequential nature within each centroid sequence is preserved due to the time series clustering while addressing class imbalance. The sequential model includes a Simple RNN layer with three units, designed to process input sequences of length 50 representing three features: eGFR values over time, age, and gender. Following this, two Dense layers are utilised: the first layer consists of 10 units with ReLU activation, while the final layer contains a single unit with a sigmoid activation function, enabling the categorisation of the output for binary classification. The model is compiled with the Adam optimiser, set at a learning rate of 0.001 and uses binary cross-entropy as the loss function.

Model training

For this study, a model was initially trained and evaluated on the Australian dataset to perform a binary classification task with the objective of predicting kidney failure based on the given features (most recent eGFR, eGFR slope, eGFR mean prior to kidney failure incidence, and age and gender). We employed explainable machine learning algorithms—Decision Tree, Random Forest—and one deep learning algorithm, namely, the Recurrent Neural Networks (RNN), for comparison. Grid search was performed to identify the optimal hyperparameter combination for each algorithm. The model's evaluation on the Australian dataset was carried out using stratified K-fold ($k = 5$) cross-validation [6]. This involved splitting the dataset into five equally sized class-stratified folds. This approach ensured that each fold was representative of the overall class distribution, mitigating class imbalance. In each iteration, the model was trained on 4 folds and validated on the remaining fold. This process was rotated through all five folds to thoroughly assess the model's performance. The performance of all classifiers was evaluated using accuracy, precision, recall, specificity, F1 score, and Receiver Operating Characteristic - Area Under the Curve (ROC-AUC) in accordance with the guidelines recommended for reporting results of clinical prediction models [18]. The model was fit with the best hyperparameters and retrained on the entire under-sampled Australian dataset. To further increase the robustness of the model, it might be better to train with different populations.

Transfer learning

One of the problems when dealing with small datasets, especially within healthcare datasets, is overfitting. Generally, overfitting occurs when a machine learning model learns not only the underlying patterns in the training data but also its noise and random fluctuations. As a result, while the model may perform exceptionally well on the training data, its performance significantly degrades on new, unseen data because it has essentially memorised the training data rather than learning to generalise from it. Considering the smallness of the datasets utilised in this study, we explored the validation of the performance of our models on an external dataset.

For this purpose, the pre-trained model on the under-sampled Australian dataset set is then evaluated on the Japanese cohort, denoted as the held-out test set. This was done by fine-tuning the model on a small subset (15 %) of the Japanese dataset with the same features before being evaluated on the remaining data. Transfer learning is a machine learning method where a model trained on a source task is repurposed as the starting point for a target task [9]. This approach negates the need to train the model from scratch in the target domain, reducing both the need for large training datasets and the time required for training [10]. By utilising a model pre-trained on the source task, transfer learning incorporates patterns from the initial dataset and addresses the common problem of insufficient training data, thereby improving a model's performance on the target dataset. This approach mitigates the potential for overfitting on the source task and improves generalisation by performing well on an unseen target dataset.

In a basic transfer learning scenario, we define a source domain D_S with data X_S and a task T_S, and a target domain D_T with data X_T and a task T_T. The goal of transfer learning is to improve the learning of the target predictive function $f_T(\cdot)$ in D_T using the knowledge from D_S and T_S.

Typically, a model is initially trained in D_S, resulting in parameters θ_S. When adapting this model to D_T, we often initialise the target model's parameters, θ_T, with θ_S (i.e., $\theta_T^{(0)} = \theta_S$), implying that the target model begins with parameters learned from the source model. During fine-tuning, θ_T is adjusted to better fit the target domain data X_T. The updated rule in an iterative training process (like gradient descent) can be represented as:

$$\theta_T^{(n+1)} = \theta_T^{(n)} - \alpha \nabla_{\theta_T} L\left(f_T\left(X_T; \theta_T^{(n)}\right), \ Y_T\right)$$

where:

- $\theta_T^{(n)}$ are the parameters at iteration n,
- α is the learning rate,
- L is the loss function that quantifies the discrepancy between the predicted outcomes,
- $f_T\left(X_T; \theta_T^{(n)}\right)$ and the actual values $Y_T$in the target domain,
- $\nabla_{\theta_T} L$ represents the gradient of the loss function with respect to the parameters θ_T, guiding the update direction.

Fig. 2 presents the comprehensive methodology, including the data selection and preprocessing, feature engineering, machine learning (ML) model development on the Australian (AU) cohort, and subsequent transfer learning on the Japanese (JP) cohort for robust cross-validation.

Results

The hyperparameter settings for each classifier are displayed in Table 2. For the decision tree, the model's hyperparameters are fine-tuned using GridSearchCV and optimised for the ROC-AUC score. The optimal configuration chosen did not incorporate class weight adjustments, and splits were determined by Gini impurity using the 'best' method, which evaluated all possible splits at a node to identify the most

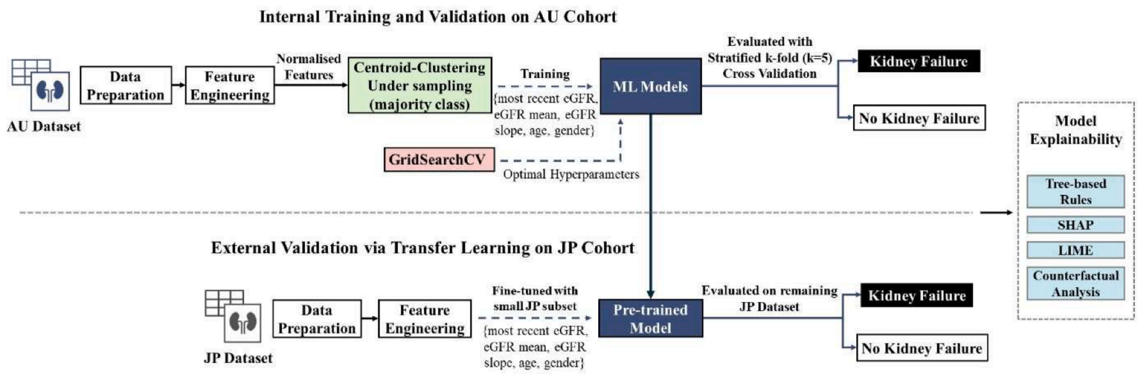

Fig. 2. The proposed CKD prediction framework evaluated on both AU and JP datasets, with internal and external validation phases.

Table 2
Hyperparameters of the algorithms.

Algorithms	Hyperparameters
Decision Tree	metric='F1′, criterion='gini', splitter='best', max_depth=None, min_samples_split=10, min_samples_leaf=4, random_state_AU=5, random_state_JP=48
Random Forest	metric='F1′, n_estimators=100, criterion='gini', max_depth=None, min_samples_split=10, min_samples_leaf=1, max_features='auto', bootstrap=False, random_state_AU=1, random_state_JP=26
RNN	metric=' F1′, Model Type='Sequential', Input Sequence Length=50, RNN units=3, Dense units Layer 1 = 10, Dense activation Layer 1=ReLU, Dense activation Layer 2='Sigmoid', optimizer='Adam', learning_rate=0.001, epochs=100, random_state_AU=0, random_state_JP=42

informative one. The tree nodes are expanded until all leaves are pure or contain fewer than min_samples_split samples, set at 10, to prevent overfitting by ensuring that each decision to divide the data is supported by a sufficient number of samples. Additionally, 'min_samples_leaf' is set to 4 and may improve the model's generalisation by requiring that there be at least a small group of data points that lead to a decision rather than decisions based on single points.

To ensure reproducibility and maintain data integrity across different cross-validation folds, an initial seed or state was set during data splitting. This guarantees, barring any noise, that simulations remain entirely deterministic based on the random seed.

The best overall performance, as measured by the ROC-AUC score, was achieved by the Random Forest algorithm (0.98, see Table 3). Nonetheless, this score was similar to the simpler and more interpretable Decision Tree model, which exhibited exceptional performance on the Australian dataset, using stratified k-fold ($k = 5$) cross-validation.

The Random Forest algorithm achieved the highest ROC-AUC score of 0.98 (see Table 3). However, the Decision Tree model, which is simpler and more interpretable, demonstrated comparable performance on the Australian dataset with stratified k-fold ($k = 5$) cross-validation.

The model, pre-trained on the under-sampled Australian dataset with optimal hyperparameters, then underwent transfer learning for evaluation on the Japanese dataset. This involved fine-tuning or calibrating the model on a small 15 % subset of the Japanese dataset, maintaining the same feature set before assessing its performance on the remainder of the data. The subset's label distribution mirrored the original dataset, ensured by stratified k-fold sampling. Similarly to the internal validation using the Australian dataset, setting the initial seed or state during data splitting plays a crucial role in influencing the outcomes. Table 4 contrasts the results of the transfer learning and evaluation process on the Japanese dataset in comparison with the Australian dataset.

The confusion matrices and ROC-AUC graphs are illustrated in Figs. 3 and 4.

The external validation on the Japanese CKD registry dataset demonstrated that the models, particularly the decision tree and random forest, maintained high predictive accuracy (ROC-AUC of 0.88 and 0.93, respectively). Transfer learning was employed by fine-tuning the pre-trained models on a small subset (15 %) of the Japanese dataset before evaluating them on the remaining data. This technique leveraged knowledge from the Australian cohort to enhance model generalization and performance on the Japanese dataset, highlighting the models' applicability across diverse populations.

Explainability

As aforementioned, we selected the decision tree for its interpretability and transparency. The models leveraged SHapley Additive exPlanations (SHAP) and Local Interpretable Model-agnostic Explanations (LIME) to provide detailed analyses of the models' decision-making

Table 3
Performance metrics with the Australian dataset.

Model	Mean F1 score	Mean ROC-AUC
Decision Tree	**0.91**	**0.94**
Random Forest	0.93	0.98
RNN	0.76	0.80

Table 4
Performance comparison of all algorithms on the Australian and Japanese datasets.

	Stratified K Fold ($K = 5$) Validation on AU Dataset (Mean and Standard Deviation)	Evaluation on JP Dataset (no fine tuning)	Evaluation on JP Dataset (15 % fine tuning)
Decision Tree			
Accuracy	0.907 ± 0.035	0.752	0.860
Precision	0.918 ± 0.047	0.597	0.691
Recall	0.895 ± 0.030	0.265	0.876
f1_score	0.906 ± 0.034	0.367	0.773
Specificity	0.918 ± 0.050	0.933	0.854
ROC-AUC	0.941 ± 0.031	0.750	0.882
Random Forest			
Accuracy	0.926 ± 0.009	0.738	0.878
Precision	0.928 ± 0.008	0.571	0.750
Recall	0.923 ± 0.021	0.148	0.826
f1_score	0.926 ± 0.010	0.235	0.786
Specificity	0.929 ± 0.009	0.958	0.897
ROC-AUC	0.976 ± 0.004	0.794	0.928
RNN			
Accuracy	0.753 ± 0.070	0.472	0.638
Precision	0.738 ± 0.074	0.331	0.422
Recall	0.801 ± 0.107	0.926	0.906
f1_score	0.764 ± 0.068	0.488	0.576
Specificity	0.705 ± 0.120	0.303	0.538
ROC-AUC	0.799 ± 0.065	0.615	0.794

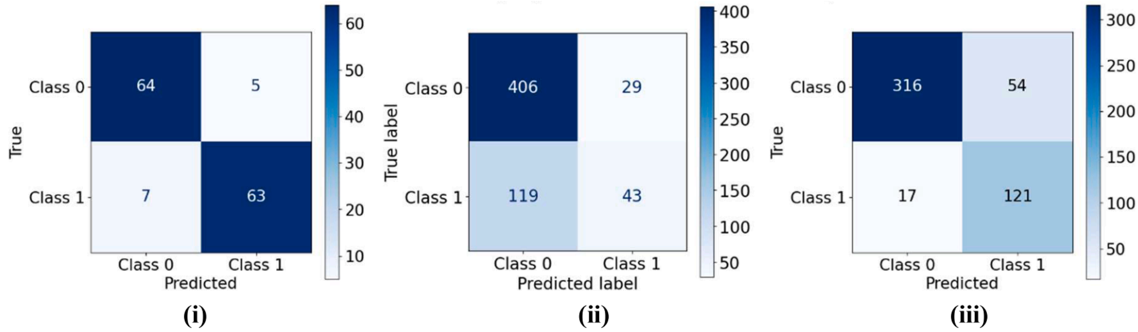

Fig. 3. Confusion Matrix for the Decision Tree i) Averaged Results across Stratified K Fold (K= 5) Validation on Australian Dataset; ii) Evaluation on Japanese Dataset (No fine tuning); iii) Evaluation on Japanese Dataset (15 % fine tuning).

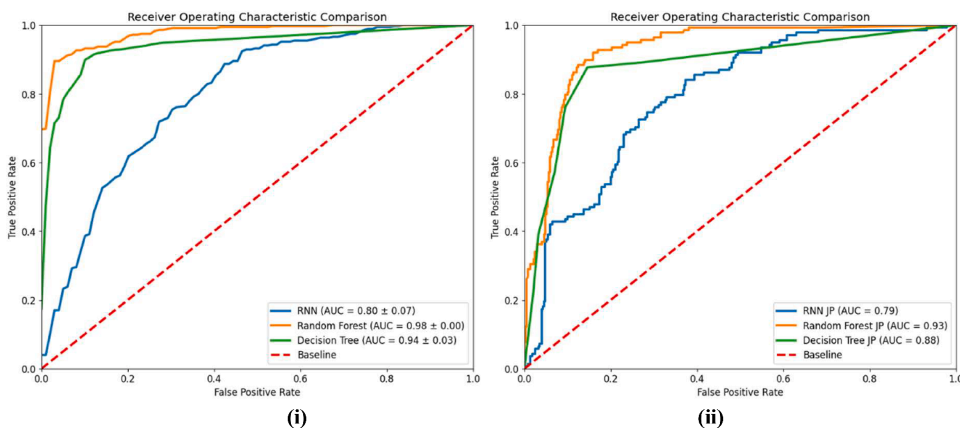

Fig. 4. i) ROC-AUC Graph for All Comparative Models: Stratified K Fold (K= 5) Validation on Australian Dataset; ii) ROC-AUC Graph for All Comparative Models: Evaluation on Japanese dataset (15 % fine tuning).

processes, enhancing their interpretability and trustworthiness in clinical settings.Key variables included age, gender, most recent estimated glomerular filtration rate (eGFR), mean eGFR, and eGFR slope over time. These variables were critical in capturing the progression of CKD and provided significant insights into the likelihood of disease progression. The importance of various features was analysed to understand the drivers of CKD progression in different populations. We found that, particularly, most recent eGFR and its slope were highly informative and improved the models' ability to predict CKD progression accurately. To find and rank the critical drivers of the prediction, the Decision Tree model was additionally trained independently on the JP dataset, along with the AU dataset, utilising stratified K-Fold validation to discern predictive differences compared to the other two approaches (see Table 5). Feature importance rankings for the AU dataset now show a significant prioritisation of eGFR (0.735), with eGFR_slope (0.172) and eGFR_mean (0.056) also contributing but to a lesser extent, followed by AGE (0.035) and SEX (0.002). For the JP dataset, the model similarly identified eGFR (0.450) as the most influential predictive feature, with eGFR_slope (0.297) and eGFR_mean (0.157) also important, whereas AGE (0.096), and SEX had no measurable impact. When the Decision Tree model was trained on the AU data and evaluated on the JP data with transfer learning, the feature importance reflected a similar hierarchy, with eGFR (0.455) remaining the most significant, followed closely by eGFR_slope (0.336) and then by eGFR_mean (0.131), AGE (0.078), and again SEX showing no discernible influence.

Table 5
Feature ranking from decision tree.

Feature	AU Importance	JP Importance	Trained AU Evaluated JP
eGFR	0.735	0.450	0.455
eGFR_slope	0.171	0.297	0.336
eGFR_mean	0.055	0.157	0.131
AGE	0.034	0.095	0.078
SEX	0.002	0.000	0.000

The prominence of the most recent eGFR in the AU cohort's feature ranking underscores its critical role in outcome prediction, emphasising its significance as a key indicator of kidney function and its broad range and variability. While still important, the most recent eGFR had a lower importance score in the Japanese dataset compared to the Australian dataset. This suggests that although current kidney function is a key predictor, its relative importance is less in the Japanese cohort. Conversely, the JP dataset's model indicates that the rate of change in eGFR, as captured by eGFR slope, is comparatively more critical for predictions than for the AU cohort, potentially due to the prevalence of negative mean eGFR slopes in the dataset, which underscores the declining kidney function's role in prognosis. Despite eGFR mean's reduced importance in the AU dataset compared to the JP dataset, it remained a relatively significant predictor, indicating nuanced differences in how eGFR trends impact prognosis across populations. The higher importance of mean eGFR in the Japanese dataset indicates that long-term kidney function trends are more influential in this population, Age held a more predictive power in the JP dataset than in the AU dataset, potentially reflecting the cohorts' demographic variations in age. The greater importance of age in the Japanese cohort suggests that older age is more strongly associated with CKD progression in this population. The negligible role of gender in both models, with no discernible influence in the JP dataset and only a marginal presence in the AU dataset, suggests that gender may have a limited direct influence on the prognostication of kidney function outcomes. When trained on the AU dataset and evaluated on the JP data, the model similarly emphasises eGFR and eGFR slope, further highlighting the consistency of these features' predictive importance across different population datasets. These variances in feature importance not only reveal intrinsic distinctions between the AU and JP datasets but also underscore the necessity for dataset-specific model tuning to enhance predictive performance in heterogeneous populations.

To better understand the underlying mechanisms driving CKD progression, particularly focusing on the most informative variables identified by the decision tree model in the Japanese cohort, such as the most recent eGFR and eGFR slope, further investigations are warranted. Differences in clinical practices, genetic factors, environmental influences, and demographic data might explain variations in disease progression. These differences could reflect variations in healthcare management, disease detection practices, and demographic factors or healthcare access and quality. Understanding these differences is crucial for developing population-specific predictive models. To achieve this, a multifaceted research approach is needed. Longitudinal studies should track eGFR changes over time to understand how fluctuations correlate with CKD outcomes. Demographic data should be analysed to understand the influence of age, gender, and other demographic variables on CKD progression. Genetic and biomolecular investigations should identify genetic variations and molecular signatures associated with eGFR decline. Environmental and lifestyle assessments should examine the impact of diet, physical activity, and other lifestyle factors on CKD progression. Healthcare practices should be compared to analyse healthcare access, frequency of medical check-ups, and treatment strategies between populations. Interventional trials should test the efficacy of drugs, lifestyle changes, or other interventions designed to stabilize eGFR. Comprehensive data integration should combine clinical, genetic, environmental, demographic, and lifestyle data for a holistic understanding of CKD progression. Such investigations will provide deeper insights into the drivers of CKD progression and inform more effective and personalized treatment strategies.

Fig. 5 provides a detailed visualisation of the decision tree model trained on 85 % of the Japanese dataset post the fine-tuning process using 15 % of the data. This classification tree represents how the model discriminates between two outcomes based on a series of feature-based decisions.

In the depicted decision tree, the primary attribute for initiating the division is eGFR, indicating its pivotal role in the model's prognostications. Subsequent to the root node, the tree utilises additional divisions based on eGFR slope and AGE, further emphasising their contribution to the classification process. Importantly, SEX does not manifest as a divisional variable at any of the principal nodes within this tree, underscoring its relatively minor significance in comparison to eGFR-related attributes and AGE. The tree structure delineates the decision-making pathway, revealing how the values of each attribute guide the prognostic outcomes. For example, the tree suggests that higher values of eGFR mean lead to distinct classification outcomes, which may correspond to differing risks associated with renal function. This decision tree provides a transparent, interpretable model that elucidates the key factors driving the classification of renal function outcomes in the Japanese dataset. It delivers a direct and comprehensible perspective of the attribute relationships and their significance, rendering the tree's logic and the impact of each variable on the outcomes readily apparent. The visual representation allows for an intuitive understanding of the model's classification logic and the significance of each variable within the dataset.

Decision tree SHAP interpretation

SHAP (Shapley Additive exPlanations) values measure the impact of each feature on the model's prediction. For decision trees, SHAP can provide an exact solution for SHAP values, representing the average contribution of a feature value to every possible prediction. Fig. 6 displays a SHAP summary plot. SHAP values explain the output of machine learning models by assigning each feature an importance value for a particular prediction. The SHAP summary plot revealed the relative importance of features in influencing a machine learning model's prediction. The feature eGFR stands out as the most influential. The features eGFR_mean, eGFR_slope and AGE also play notable roles, whereas the SEX feature appears to have a more subdued impact on the model's predictions overall. The visualisation in Fig. 6 underscores the importance of understanding individual feature contributions to enhance the model's interpretability and trustworthiness. Then, across each feature, each dot represents the impact of a feature value on the model's prediction for a single patient, with the position on the x-axis indicating the magnitude and direction of the impact. In the plot for the positive class, it is observed that higher eGFR values tend to decrease the risk of kidney failure (blue dots on the left side), while lower values increase the risk (pink dots on the right side). This aligns with medical understanding, as eGFR is a critical indicator of kidney function.

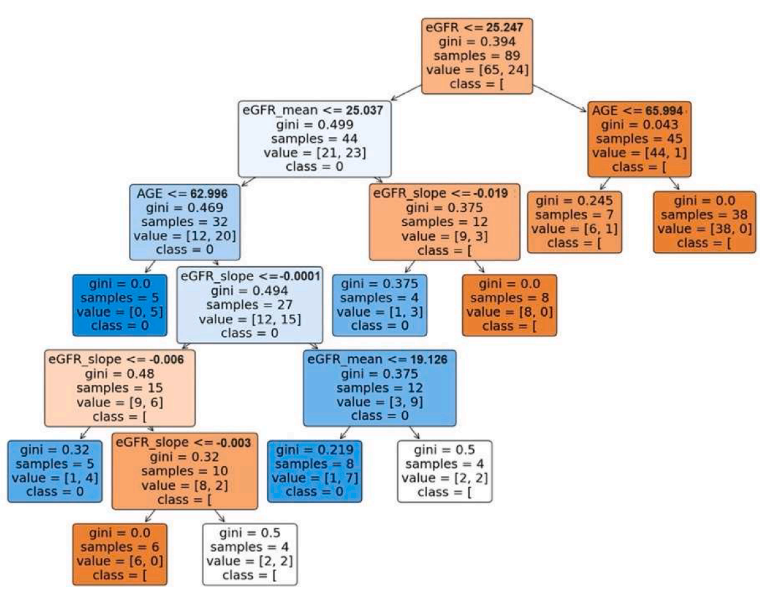

Fig. 5. Decision Tree Rules (Cross-validation on Japanese dataset).

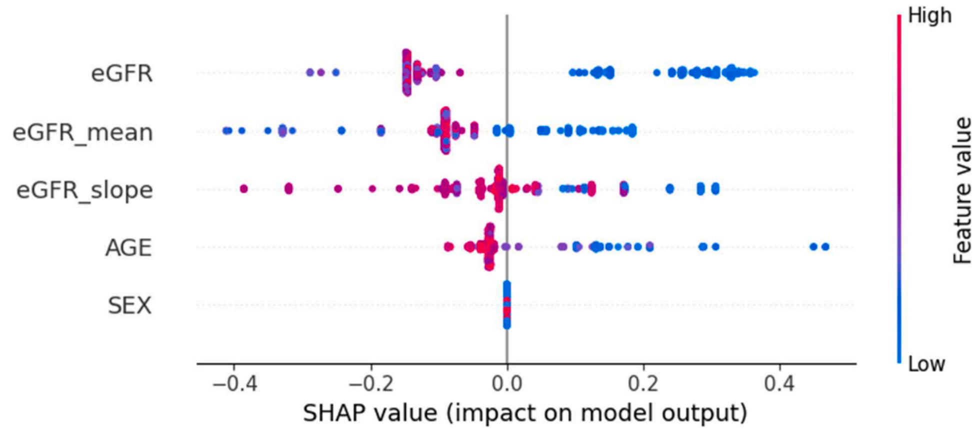

Fig. 6. SHAP summary plot illustrating the global influence of input variables on the predictive performance of the Decision Tree model (cross-validation on Japanese dataset), using SHAP values.

Decision tree lime interpretation

Additionally, LIME analysis was applied to analyse and interpret the prediction for a particular instance. The LIME results offer insights into the local behaviour of a machine-learning model for a specific prediction. This analysis was illustrated by examining an individual patient, as depicted in Fig. 7. The interpretive figure showcases how specific features sway the classification towards or against Kidney Failure, with orange indicating a positive influence and blue a negative one. The figure's left side presents the prediction probabilities, and the central portion ranks the most influential features by bar length. For this instance, the model predicts "Kidney Failure" with absolute certainty. In the figure's explanation, the eGFR mean (15.36 mL/min/1.73m^2) feature heavily favours a Kidney Failure outcome, followed by eGFR (15.33 mL/min/1.73m^2) and AGE (27 years). The eGFR slope (−0.0013 mL/min/1.73m^2/day) has a very minor influence, and SEX (male) has no impact on this prediction. This aligns with the clinical understanding that lower eGFR values are indicative of poorer kidney function, and the model has learned to heavily weigh these features when predicting Kidney Failure. The actual label for this case is also "Kidney Failure," confirming the model's prediction. In summary, the model's prediction of "Kidney Failure" is strongly guided by the eGFR mean, eGFR and AGE features. LIME clarifies these individual predictions, contrasting them with the broader, often opaque model behaviours. LIME's strength lies in explaining individual predictions, rather than global model behaviour, by creating a local surrogate model that approximates the behaviour of the complex model around the vicinity of a particular instance. The surrogate model, often a linear model, is interpretable because it weights the input features in a way that can be easily understood. LIME can help understand the rationale behind specific decisions made by complex models. This contrasts with more complex models, like deep neural networks or ensemble methods, which might be considered "black boxes" due to their intricate internal workings that are not readily interpretable.

Counterfactual analysis

Counterfactual analysis is an interpretability technique for understanding model predictions by exploring "what if" scenarios and querying changes required to impact the predicted outcomes. It involves generating counterfactual examples: slight alterations to the input features of a given instance that would lead to a different prediction outcome. This analysis is valuable for interpretability and explainability in machine learning, as it provides actionable insights into how feature values influence predictions, enhancing transparency and trust in complex models. DiCE (Diverse Counterfactual Explanations) is an interpretability framework designed to help understand ML model predictions by generating counterfactual explanations [19]. These hypothetical scenarios show minimal changes needed in the input features of a data instance to achieve a different prediction outcome. Generating counterfactuals with DiCE involves finding instances in the feature space close to the original instance, which results in a different prediction. In the case of patient UID 67, which was predicted to have Kidney Failure, DiCE was used to find what changes could be made to this instance's features to change the model's prediction to Non-Kidney Failure. A counterfactual example that resulted in this was changing the eGFR mean from 15.396 mL/min/1.73m^2 to 25.483 mL/min/1.73m^2, implying that increasing the average eGFR value slightly could also contribute to a Non-Kidney Failure prediction. Other features remain unchanged in this counterfactual example, highlighting that altering the eGFR alone, in this case, could lead to the desired prediction change. The counterfactuals suggest that small adjustments to eGFR, eGFR mean, and eGFR slope could lead to a prediction of non-Kidney failure for this patient. These changes are hypothetical and need to be interpreted within the clinical context. For example, an increase in eGFR values generally indicates better kidney function, which aligns with the counterfactual prediction of non-Kidney failure. It is important to note that these counterfactuals do not imply causation but provide potential scenarios that could lead to a different prediction, which can be useful

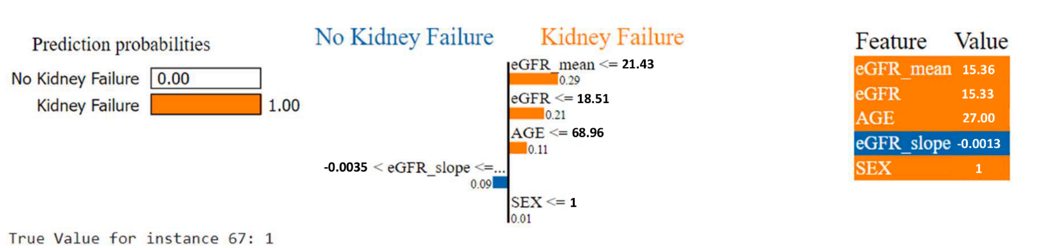

Fig. 7. LIME plot illustrating the local influence of input variables on the predictive performance of the Decision Tree model (cross-validation on Japanese dataset) for a particular patient instance.

for understanding model behaviour and sensitivities. The effectiveness and plausibility of these counterfactuals depend on the accuracy and reliability of the model and the representativeness of the data it was trained on. In healthcare applications, any insights or actions derived from such analyses should be evaluated and validated by medical professionals. This analysis could extend to more personalised and alterable variables such as BMI, smoking and other lifestyle factors.

Discussion

Early identification of individuals at risk of progressing to kidney failure is crucial, as it allows for targeted interventions that can significantly alter the course of the disease [5].

Although CKD progresses through each stage sequentially, it is often clinically silent, meaning that most affected individuals are unaware of their condition [5,7]. As a result, many are diagnosed at mid to late stages, necessitating urgent treatment. The early methods for predicting CKD progression involve using statistical methods - the most sophisticated currently being the KFRE [11]. Although this equation provides great insight into the kidney failure detection, it is a static equation which focus only on the current values and does not take the time-series changes into consideration for kidney failure prediction. The advent of AI has brought about a revolutionary shift in data handling and processing [20]. Although the fundamental calculations remain consistent, the scale and predictive capabilities have entered a completely new realm. Traditional methods can be both expensive and time-consuming, leaving greater margin for human error; thus, machine learning offers promising solutions for these reasons. Machine learning has made it feasible to extract meaningful insights from what would otherwise be inscrutable data, capturing underlying patterns and fluctuations. Therefore, machine learning is better suited than traditional statistical models when dealing with non-linear data.

The current literature suggests a keen interest in the development of effective clinical utility prognostic model with the ability to predict kidney failure [10,16]. To achieve this, larger and more time-series data is being used for model training. Using machine learning for time-based variables rather than traditional regression models allows it to capture complex non-linear relationships. This cost-efficient, generalisable tool helps predict progressive CKD earlier across all disease stages compared to previous kidney failure prediction models.

The literature has also emphasised that the explainability of the CKD predictive model is necessary in a clinical environment [21]. Training models from time-series data enable more complex, non-linear relationships between variables without resorting to 'black-box' methods such as deep neural networks where explainability is often a major difficulty. A deeper understanding of CKD over time can allow professionals in clinical settings to study disease evolution, paving the path to more personalised and specific medicine approaches. Various studies have taken the initiative and have expressed the need to build models based on time-varying data set [10,16]. It is also observed that studies have started to adopt more interpretable methods [14], recognising that the clinical practice requires a certain degree of interpretability for practical implementation. Our study has employed techniques such as LIME and SHAP, which provide a detailed analysis of the model's decision-making process and rationale, offering insights at the local level (for an individual patient) and at the global level (across the entire dataset). This analysis helps in understanding the predictive behaviour of the model, including the identification of key features and impact on predictions, thereby offering a comprehensive view of how the model arrives at its conclusions in a wide range of scenarios. Utilizing counterfactual analysis, we further explore 'what-if' scenarios, examining the impact of the smallest changes in input variables on the model's predicted risk of kidney failure.

Our retrospective cohort study demonstrates the potential utility of explainable machine learning approaches in enhancing the prediction of chronic kidney disease (CKD) progression. The results highlight exceptional predictive accuracy, including AUC-ROC scores of 0.94 and 0.98 for the decision tree and random forest models, respectively, in the Australian dataset. Following a transfer learning approach to enable validation in the smaller Japanese dataset of just 597 patients, AUC-ROC scores of 0.88 and 0.93 were achieved for the decision tree and random forest models, respectively. The high level of performance exhibited by these interpretable models underscores their ability to predict CKD outcomes even on limited sample sizes, while also providing clinical interpretability. The robust AUC-ROC values in our study are consistent with previous studies demonstrating the capabilities of machine learning for prognostic modelling in CKD. Our finding of comparable high accuracy for the random forest model aligns with these results and establishes machine learning techniques as valuable tools for risk-stratifying CKD patients.

Furthermore, the use of transfer learning, though novel in the CKD domain, enabled the validation of our models on the smaller Japanese dataset while retaining accuracy. Transfer learning involves transferring knowledge gained from training a model on a source dataset to improve performance on a different target dataset. This technique has shown tremendous promise for medical imaging, genomic analyses, and other biomedical applications with limited data [22]. By leveraging the Australian discovery cohort to pre-train, the decision tree and random forest models before fine-tuning and evaluation on the Japanese dataset, we mitigated overfitting issues and demonstrated wider applicability across ethnically diverse populations.

The feature importance analysis also provides clinical interpretability by revealing the relative influence of patient age, most recent eGFR, eGFR slope, and mean eGFR on outcome predictions by the models. Understanding the impact of these variables can inform targeted early interventions to mitigate risk. For example, increased monitoring and renoprotective therapies could be implemented for patients exhibiting a sharply declining eGFR over time. Previous studies have similarly called for explainable and interpretable machine learning models in CKD to enable actionable insights into patient trajectories [5,6].

Several limitations temper the results and warrant discussion. The modest sample sizes, though partly addressed via transfer learning, necessitate validation in considerably larger independent cohorts. A large and geographically diverse outpatient cohort would have strengthened its overall generalizability and usability. Additionally, the consistency of the models over longer follow-up was not evaluated. Furthermore, seamless integration with clinical workflow, including EHR systems, would be essential to enable pragmatic implementation. Finally, additional renal and cardiometabolic parameters could be incorporated to enhance risk stratification capabilities. In particular, while transfer learning improved predictive accuracy, it introduced several limitations and challenges. One major issue is domain shift: the Australian and Japanese populations may have differences in disease aetiology, lifestyle, healthcare access, practices, and genetic factors influencing CKD progression, potentially impacting model performance. Fine-tuning on a small subset of the Japanese dataset risks overfitting and may not capture the full variability of the target population, limiting generalizability. Ensuring the model's interpretability is crucial for clinical adoption and trust. Addressing these challenges requires extensive cross-validation to assess performance across diverse subgroups, employing interpretable machine learning techniques like SHAP and LIME for transparency, and ensuring compliance with ethical standards and thorough validation to avoid bias. Engaging multidisciplinary teams of domain experts, clinicians, and data scientists ensures clinical relevance and robustness.

Our findings provide preliminary evidence supporting the integration of novel prognostic variables and interpretable machine learning techniques to predict early CKD progression. Some machine-learning models rely primarily on novel kidney disease biomarkers [23], while our models use readily obtainable laboratory information.The high predictive accuracy and transportability exhibited by the decision tree and random forest models position them as potentially valuable augmentations to existing tools like KFRE. Following external validation, such models could assist clinicians by identifying patients at high-risk of progression and who would likely benefit from close monitoring and treatment escalation. Future studies should implement these methods prospectively over extended periods in diverse CKD populations.

We propose integrating the developed CKD progression models into existing electronic health record (EHR) systems. This integration would involve creating a dedicated module within the EHR that automatically pulls relevant patient data to generate real-time risk predictions. The module would present these predictions to clinicians during patient encounters, along with key contributing factors and suggested interventions. To ensure smooth adoption, we recommend a phased implementation approach, starting with a pilot study in select nephrology clinics. This pilot would assess the model's impact on clinical decision-making, patient outcomes, and workflow efficiency. Comprehensive training programs for healthcare providers would be crucial, focusing on interpreting model outputs and incorporating them into patient care plans. Additionally, establishing a feedback mechanism would allow for continuous refinement of the model based on real-world performance and clinician input. By carefully integrating these predictive tools into clinical workflows, we can enhance early detection and management of CKD progression while minimizing disruption to existing practices.

It must also be noted that use of AI in clinical decision-making raises ethical concerns and potential biases [20]. Privacy and accountability further complicate their implementation. Addressing these concerns requires appropriate governance including interdisciplinary collaboration to develop guidelines for responsible use of the models, ensuring ongoing evaluation and adaptation to maintain fairness and effectiveness.

Conclusion

In conclusion, this study has successfully developed explainable machine learning models utilising routinely collected pathology data to accurately predict the trajectory of CKD. By leveraging key variables such as age, gender, and various aspects of estimated eGFR, the models demonstrated exceptional predictive accuracy, particularly in identifying the risk of progression to kidney failure. Internal validation on an Australian cohort showcased high performance, with area under the receiver operating characteristic curve (ROC-AUC) values of 0.94 and 0.98 for decision tree and random forest algorithms, respectively. Furthermore, external validation on a Japanese dataset, followed by transfer learning techniques to enhance generalisation, yielded respectable accuracy scores, indicating the potential applicability of the models across diverse populations. Notably, the identified variables align with the underlying pathophysiology of CKD, emphasising the clinical relevance of the developed models. Additionally, interpretable ML techniques such as SHAP and LIME for further analysis provided insights into model behaviour at both local and global levels. These methodologies contributed coherent and rational explanations, thereby enhancing the model's overall transparency. While further validation in larger multi-ethnic cohorts is warranted to strengthen generalizability, the study's findings underscore the promise of accurate predictive modelling in identifying high-risk CKD patients for personalised disease management interventions. Research on CKD progression should focus on key areas identified in the Japanese cohort study, including eGFR dynamics, hemodynamic factors, RAAS, inflammatory pathways, podocyte biology, and metabolic influences. Additional investigation into genetic and epigenetic factors, novel biomarkers, and environmental influences is crucial for a comprehensive understanding of disease mechanisms. This approach may identify new intervention targets and develop personalized strategies to slow or halt CKD progression.

Data availability

The code for the machine learning models is available at https://github.com/roysup/Chronic-Kidney-Disease

CRediT authorship contribution statement

Sandeep Reddy: Writing – review & editing, Writing – original draft, Supervision, Methodology, Formal analysis, Conceptualization. **Supriya Roy:** Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Kay Weng Choy:** Writing – review & editing, Data curation, Conceptualization. **Sourav Sharma:** Formal analysis, Data curation. **Karen M Dwyer:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Chaitanya Manapragada:** Methodology. **Zane Miller:** Investigation. **Joy Cheon:** Investigation. **Bahareh Nakisa:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.cmpbup.2024.100160.

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